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An analytical cyclo-steady-state model for inter-rifting deformation at magmatic rifts

Iván Navarrete[1]
Jean-Arthur Olive[1]
[1] *Laboratoire de Géologie, École Normale Supérieure, PSL Research University, Paris, France*

SUMMARY
Inter-rifting deformation (the strain that accumulates between discrete magmatic episodes) remains the least-constrained phase of the rifting cycle. Existing kinematic frameworks represent the rift as a continuously opening dislocation in an elastic half-space, an assumption incompatible with both the known rheological structure of magmatic rifts and far-field geodetic constraints. We develop a fully analytical model for the surface fields generated by a periodic sequence of instantaneous dike intrusions in an elastic plate overlying a Newtonian viscous half-space. The governing equation reduces, in the Fourier domain, to a fractional (half-Laplacian) diffusion equation parameterized by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$, where E, H, and ν are the Young's modulus, thickness, and Poisson's ratio of the brittle layer, and η is the asthenospheric viscosity. In the limit of many repeated intrusions the system reaches a Cyclo-Steady State (CSS) with a closed-form expression through the Digamma and log-Gamma functions, conserving far-field plate divergence by construction. We extend the model to a Maxwell visco-elastic asthenosphere via the elastic–visco-elastic correspondence principle, recovering two physically distinct relaxation regimes separated by the Maxwell relaxation time τ_R: near-field elastic rebound for $t \lesssim \tau_R$ and far-field viscous diffusion for $t \gg \tau_R$. At Krafla (Foulger et al., 1992), the viscous step response is favoured over diffusion models ($\Delta\mathrm{AIC} = -3.3$) and yields $\eta \approx 4\times10^{18}$ Pa s ($H = 10$ km, $E = 40$ GPa); at Asal (Vigny et al. 2007), independent physical constraints favour the same formulation and give $\eta \approx 2 \times 10^{19}$ Pa s, a factor of five higher than at Krafla, reflecting the anomalous sub-lithospheric heat flux beneath Iceland. The CSS framework provides a mechanically self-consistent description of inter-rifting deformation at any magmatic rift system, yielding viscosity estimates from a single geodetic velocity profile without auxiliary rheological assumptions.

Key words: continental rifting; inter-rifting deformation; fractional diffusion; visco-elastic relaxation; GPS geodesy; Iceland

1 INTRODUCTION

Continental rifting drives the transition from stable cratonic lithosphere to active oceanic spreading through rifting cycles that span a broad range of spatial and temporal scales (Corti 2009; Ebinger et al. 2013). These cycles are conventionally partitioned into three phases: elastic strain release through dike emplacement and normal faulting (Grandin et al. 2011; Maccaferri et al. 2016); post-diking visco-elastic relaxation, which decays on timescales of years to decades (Foulger et al. 1992; Nooner et al. 2009); and an inter-rifting phase during which the plates diverge at a secular rate set by large-scale mantle dynamics (Saria et al. 2014; Stamps et al. 2021). Geodetic campaigns in Iceland (Heki et al. 1993; Hofton et al. 1995) and the Afar (Vigny et al. 2007; Nooner et al. 2009; Grandin et al. 2011) have documented the first two phases in considerable detail. The mechanisms governing the build-up of extensional strain between discrete events, however, remain the least-constrained aspect of the rifting cycle.

The interseismic strain accumulation in rifts is poorly constrained (Ebinger et al. 2013). Modelling the inter-rifting phase is conventionally done by transplanting kinematic frameworks developed for inter-seismic earthquake cycle. The standard approach to this problem borrows the backslip model developed for the interseismic earthquake cycle (Savage & Prescott 1978). The rift is assumed locked to depth H, below which it slips continuously, generating the basal shear forces needed to reproduce observed velocity profiles. Alternative formulations represent the rift as a dislocation in an elastic half-space that opens at a depth-dependent

rate (LaFemina et al. 2005; Konca et al. 2018), or prescribe a basal shear traction to mimic far-field plate forcing (Ali et al. 2014). These models share a fundamental physical inconsistency: they ignore the rheological transition at the brittle–ductile boundary of magmatic rifts, where high heat flow and partial melt weaken the lower crust and uppermost mantle (Prodehl et al. 1997; Plasman et al. 2017). Geodetic and seismic constraints place this transition at 5–15 km depth in Iceland and the Afar (Stevens et al. 2021; Weinstein et al. 2017), implying visco-elastic relaxation times incompatible with the purely elastic response assumed by backslip models. A continuously opening dislocation is equally problematic: dike intrusions are finite, discrete events, and the geodetic record shows that each episode is followed by a localized transient decaying within a few years (Foulger et al. 1992; Reiss et al. 2022).

Efforts to model the post-diking transient have confronted the same boundary-condition obstacle from two directions. Analytical approaches, exemplified by the diffusion model of Foulger et al. (1992), reproduce the broad deformation pattern following individual intrusions but require an a priori specification of the viscous layer thickness as a free parameter. Numerical visco-elastic simulations (Hofton et al. 1995; Hofton & Foulger 1996; Nooner et al. 2009) face an equivalent difficulty: in any finite domain, velocities must vanish at the lateral boundaries, a condition irreconcilable with the steady plate divergence observed in global reference frames (Fernandes et al. 2004; Altamimi et al. 2016). The result is a kinematic gap between the transient post-diking signal and the secular strain rates of global plate motion models (Kreemer et al. 2014; Stamps et al. 2018).

The theoretical foundations laid for cyclic strike-slip fault deformation (Nur & Mavko 1974; Savage & Prescott 1978; Spence & Turcotte 1979; Cohen & Kramer 1984) suggest a different perspective. Inter-rifting deformation is not a distinct physical process but the emergent superposition of visco-elastic responses to an infinite history of discrete dike intrusions. The apparent steadiness of inter-rifting velocities arises naturally from the time-averaging of decaying transients, with no need for a continuously creeping source.

We develop an analytical model for an elastic crust overlying a viscous half-space (Hetland 2006) in which dike intrusions recur periodically at interval T and each open the rift by $2b$, where b is the dike half-opening. The governing integro-differential equation reduces, after a Fourier transform, to a fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993) parameterized by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$. In the limit of many repeated intrusions, the system reaches a Cyclo-Steady State (CSS) expressed exactly through the Digamma and log-Gamma functions, which satisfies the far-field plate divergence constraint by construction. We then extend the model to a Maxwell visco-elastic asthenosphere via the elastic–visco-elastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010), recovering two physically distinct relaxation regimes separated by the Maxwell timescale $\tau_R = \eta/\mu_m$. We apply both models to two geodetic datasets: horizontal displacement profiles from the Krafla rift, Iceland, recorded 8–11 years after the 1975–1984 rifting sequence (Foulger et al. 1992), and a 25-year baseline time series from the Asal rift, Djibouti (Vigny et al. 2007). Together, they constrain asthenospheric viscosity across settings separated by a factor of five, and show that decadal post-rifting transient and secular plate divergence are the short- and long-term expression of a singly underlying process. and secular plate divergence.

2 MODEL

2.1 Physical setup and governing equations

The model represents the lithosphere as a coupled mechanical system: a thin elastic plate of thickness H (the seismogenic crust) overlying a Newtonian viscous half-space of viscosity η representing the asthenosphere (e.g., Hetland & Hager 2005; Hetland 2006). The horizontal coordinate x is perpendicular to the rift axis; the vertical coordinate z is oriented downward, with the free surface at $z = 0$ and the crust–mantle interface at $z = H$. Under plane-strain conditions with negligible vertical stress ($\sigma_{zz} \approx 0$), the depth-averaged horizontal stress in the elastic layer is related to the horizontal surface displacement $u(x, t)$ through Hooke's law:

$$\bar{\sigma}_{xx} = \frac{E}{1-\nu^2}\frac{\partial u}{\partial x}, \tag{1}$$

where E is Young's modulus and ν is Poisson's ratio. The factor $(1-\nu^2)^{-1}$ accounts for the lateral confinement of plane strain. Mechanical equilibrium of the plate requires the horizontal gradient of the depth-averaged stress to be balanced by the shear traction exerted at its base by the underlying viscous mantle:

$$H\frac{\partial \bar{\sigma}_{xx}}{\partial x} + \sigma_{xz}^{\mathrm{BOT}} = 0. \tag{2}$$

The rheological character of the asthenosphere enters through the constitutive relation for the basal traction $\sigma_{xz}^{\mathrm{BOT}}$. For a Newtonian viscous half-space, the shear stress at x depends on the surface velocity at *all* other points through the non-local Stokes kernel (Savage & Prescott 1978):

$$\sigma_{xz}^{\mathrm{BOT}}(x,t) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}(x',t)\,dx', \tag{3}$$

interpreted as Hadamard finite part. This non-local coupling reflects the long-range character of viscous flow in the half-space: a velocity perturbation at x' decays as $(x-x')^{-2}$ rather than being confined to a finite diffusion radius. In magmatically active rifts, where high heat flow and partial melt weaken the lower crust (Holtzman 2016; Plasman et al. 2017), this coupling operates across the entire plate boundary system. Substituting (1) into (2) and using (3) yields the governing integro-differential equation for u:

$$\frac{EH}{1-\nu^2}\frac{\partial^2 u}{\partial x^2} + \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}(x',t)\,dx' = 0. \tag{4}$$

Dike intrusions enter as initial conditions: each event opens the rift instantaneously by $2b$, displacing the right side by $+b$ and the left by $-b$ (Figure 1).

2.2 Fourier reduction and the characteristic velocity

Equation (4) is non-local in x and not tractable in physical space. A spatial Fourier transform resolves this, because

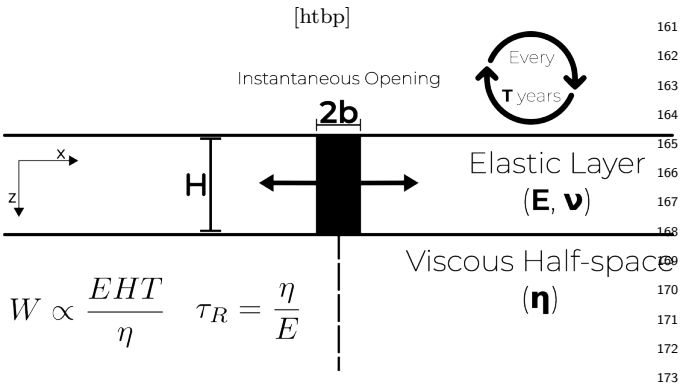


Figure 1. Model geometry. The brittle elastic plate (thickness H, Young's modulus E, Poisson's ratio ν) overlies a Newtonian viscous half-space (viscosity η). Dike intrusions open the rift by $2b$ at intervals T; the surface displacement $u(x,t)$ is positive away from the rift axis. The parameter $a = EH/[2\eta(1-\nu^2)]$ sets the characteristic velocity that couples elastic and viscous deformation.

the Stokes convolution diagonalizes in wavenumber space. Defining the transform pair

$$\hat{u}(k,t) = \int_{-\infty}^{+\infty} u(x,t)e^{-ikx}\,dx, \qquad u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \hat{u}(k,t)e^{ikx}\,dk,$$

the transform of $\partial^2 u/\partial x^2$ is $-k^2\hat{u}$. The transform of the Stokes kernel convolution is the classical result (Savage & Prescott 1978):

$$\mathcal{F}\left\{\int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}\,dx'\right\} = -2\eta|k|\,\frac{\partial \hat{u}}{\partial t}. \tag{5}$$

Applying the Fourier transform to (4) and using (5):

$$-\frac{EH}{1-\nu^2}k^2\hat{u} - 2\eta|k|\frac{\partial \hat{u}}{\partial t} = 0. \tag{6}$$

Rearranging and introducing the characteristic velocity

$$\boxed{a := \frac{EH}{2\eta(1-\nu^2)}}, \tag{7}$$

equation (6) simplifies to a decoupled first-order ODE for each wavenumber k:

$$\frac{\partial \hat{u}}{\partial t} = -a\,|k|\,\hat{u}. \tag{8}$$

The parameter a encodes the full elastic–viscous coupling of the system: large a (stiff crust, low viscosity) implies rapid redistribution of post-diking strain over broad spatial scales; small a implies slow, locally confined relaxation. In physical space, the operator $|k|$ corresponds to the fractional half-Laplacian $(-\Delta)^{1/2}$, so (8) is equivalently written as the fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993; Samko et al. 1993)

$$\frac{\partial u}{\partial t} = -a\,(-\Delta)^{1/2}u. \tag{9}$$

This order is not a modelling choice. The fractional Laplacian admits a representation as a Dirichlet-to-Neumann map for a harmonic extension problem into an auxiliary half-space, weighted by y^{1-2s}; the weight vanishes, and the extension becomes the ordinary Laplace equation, only at $s = 1/2$ (Caffarelli & Silvestre 2007; Daoud & Laamri 2022). That is the same half-space potential problem generating the Stokes kernel itself, so the half-order non-locality in (9) is forced by the geometry of the elastic-plate–viscous-half-space coupling, not fitted to the data. Solving the full two-dimensional Stokes flow beneath the plate, governed by the biharmonic rather than the harmonic equation, still recovers the same interface operator $-2\eta(-\Delta)^{1/2}$ exactly once the interface stress is extracted.

This contrasts with classical Fickian diffusion: the half-order operator, a direct consequence of the non-local Stokes kernel, produces Lorentzian rather than Gaussian spatial profiles, with tails that decay as $1/x$ rather than exponentially. The solution of (8) for arbitrary initial data is:

$$\hat{u}(k,t) = \hat{u}(k,0)\,e^{-a|k|t}. \tag{10}$$

2.3 Step response to a single dike intrusion

The simplest building block of the model is the deformation produced by a single instantaneous dike intrusion at $t = 0$. A symmetric opening of $2b$ displaces the right side of the rift by $+b$ and the left by $-b$. Exploiting the antisymmetry of the problem, the initial condition is a Heaviside displacement:

$$u(x,0^+) = b\,\mathcal{H}(-x). \tag{11}$$

Applying the Fourier transform to (11) gives $\hat{u}(k,0) = b\int_{-\infty}^{0} e^{-ikx}\,dx$, which for real k is treated in the distributional sense by adding a convergence regulator $\varepsilon > 0$:

$$\int_{-\infty}^{0} e^{-ikx}\,dx = \lim_{\varepsilon\to 0}\int_{-\infty}^{0} e^{(-ik+\varepsilon)x}\,dx = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon - ik}.$$

By the Sokhotski–Plemelj theorem (Ablowitz & Fokas 2003),

$$\lim_{\varepsilon\to 0^+}\frac{1}{x \pm i\varepsilon} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x), \tag{12}$$

where $\mathcal{P}$ denotes the Cauchy principal value. Multiplying numerator and denominator by i recasts the regulator as $+i\varepsilon$,

$$\frac{1}{\varepsilon - ik} = \frac{i}{i\varepsilon + k} = \frac{i}{k + i\varepsilon},$$

so that (12) applies directly with $x = k$ and the upper sign:

$$\begin{aligned}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon - ik} &= \lim_{\varepsilon\to 0^+}\frac{i}{k+i\varepsilon}\\ &= i\left[\mathcal{P}\left(\frac{1}{k}\right) - i\pi\delta(k)\right]\\ &= \pi\delta(k) + \frac{i}{k},\end{aligned} \tag{13}$$

where the principal-value symbol on $1/k$ is dropped from here on, as is standard, since it is understood implicitly whenever $1/k$ appears next to a $\delta(k)$ term. Hence the initial Fourier spectrum is:

$$\hat{u}(k,0) = b\pi\delta(k) + \frac{ib}{k}. \tag{14}$$

Substituting (14) into (10) and noting that $\delta(k)$ is active only where $e^{-a|k|t} = 1$:

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k}\,e^{-a|k|t}. \tag{15}$$

The inverse Fourier transform splits into two contributions, $u(x,t) = I_1 + I_2$. The delta-function term gives a constant background $I_1 = b/2$; for I_2, the parity of $\sin(kx)/k$ yields:

$$I_2 = -\frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\, e^{-akt}\, dk. \tag{16}$$

This is a standard Dirichlet-type integral. Applying the complementary arctangent identity $\arctan(z) + \arctan(1/z) = \pi/2$, the step displacement is:

$$\boxed{u^{\mathrm{STEP}}(x,t) = \frac{b}{\pi}\arctan\left(\frac{at}{x}\right).} \tag{17}$$

Differentiating with respect to t, the step velocity is a Lorentzian profile that broadens in time:

$$v^{\mathrm{STEP}}(x,t) = \frac{ba}{\pi}\,\frac{x}{x^2 + a^2t^2}. \tag{18}$$

Both expressions satisfy the correct limits: $u^{\mathrm{STEP}} \to b/2$ and $v^{\mathrm{STEP}} \to 0$ as $t \to \infty$, consistent with far-field plate divergence. The velocity peak migrates outward as at, with amplitude decaying as $1/t$.

2.4 Cyclic accumulation and the Cyclo-Steady State

2.4.1 Finite sum over past intrusions

A rift that has experienced $n = \lfloor t/T \rfloor$ dike intrusions at regular intervals of T years accumulates displacement by superposition of individual step responses:

$$u(x,t) = \frac{b}{\pi}\sum_{j=1}^{n}\arctan\left(\frac{a(t-jT)}{x}\right). \tag{19}$$

Differentiating with respect to t, the corresponding velocity field is:

$$v(x,t) = \frac{ba}{\pi}\sum_{j=1}^{n}\frac{x}{x^2 + a^2(t-jT)^2}. \tag{20}$$

2.4.2 The CSS limit and Digamma representation

After many cycles the system enters a periodic state, the rift analog of the cyclic steady state described for the interseismic earthquake cycle (Thatcher & Rundle 1984; Hetland & Hager 2006). Introducing $\tau \in [0, T)$ as the time elapsed since the most recent intrusion, the Cyclo-Steady State (CSS) velocity is obtained by extending (20) to an infinite history:

$$v_{\mathrm{CSS}}(x,\tau) = \frac{ba}{\pi}\sum_{j=1}^{\infty}\frac{x}{x^2 + a^2(\tau-jT)^2}. \tag{21}$$

Because this series converges absolutely, it admits a closed-form expression via the Digamma function $\psi(z)$. Using the partial-fraction identity

$$\frac{x}{x^2 + a^2(\tau-jT)^2} = \frac{1}{2i}\left[\frac{1}{a(\tau-jT) - ix} - \frac{1}{a(\tau-jT) + ix}\right],$$

factoring $-aT$ from each denominator and substituting into (21), the double sum takes the form of differences of Digamma series. Using the representation $\psi(z+1) = -\gamma + \sum_{n=1}^{\infty}(1/n - 1/(n+z))$, the Euler–Mascheroni constant γ and the $1/n$ terms cancel when the two conjugate series are differenced. With $Z = \tau/T + ix/(aT)$ and the identity $\psi(Z) - \psi(\bar{Z}) = 2i\,\mathrm{Im}\{\psi(Z)\}$, the CSS velocity collapses to:

$$\boxed{v_{\mathrm{CSS}}(x,\tau) = \frac{2b}{\pi T}\,\mathrm{Im}\left\{\psi\left(\frac{\tau}{T} + i\,\frac{x}{aT}\right)\right\}.} \tag{22}$$

2.4.3 CSS displacement and far-field conservation

Integrating (22) over τ, and using the antiderivative $d[\ln\Gamma(z)]/dz = \psi(z)$, the CSS displacement is:

$$\boxed{u_{\mathrm{CSS}}(x,\tau) = \frac{2b}{\pi}\,\mathrm{Im}\left\{\ln\Gamma\left(\frac{\tau}{T} + i\,\frac{x}{aT}\right) - \ln\Gamma\left(i\,\frac{x}{aT}\right)\right\}.} \tag{23}$$

Over one complete cycle, the cumulative displacement at any fixed $x > 0$ equals the dike half-opening:

$$\int_0^T v_{\mathrm{CSS}}(x,\tau)\, d\tau = b, \qquad \forall\, x > 0. \tag{24}$$

This is not imposed as a boundary condition but follows analytically from the Digamma representation: $\Gamma(1+iq) = iq\,\Gamma(iq)$ implies $\mathrm{Im}\{\ln\Gamma(1+iq) - \ln\Gamma(iq)\} = \pi/2$, giving the integral b. The CSS thus reconciles high-frequency magmatic pulses with secular divergence without unphysical boundary conditions, satisfying the far-field constraint of global plate motion models (Saria et al. 2014; Stamps et al. 2021) by construction.

The spatial width of the CSS deformation zone scales as $aT \propto EHT/\eta$. Because the CSS depends on the rift parameters (b, T) only through the product aT, geodetic profiles of inter-rifting deformation directly constrain the ratio EH/η, providing a new route to lithospheric characterization in systems where T can be independently estimated (Figure 2a). The finite-sum approximation to the CSS converges to within the measurement noise of a typical geodetic network after approximately 10 rifting cycles (Figure 2b); for systems with fewer events on record, the CSS should be replaced by the explicit sum in (19), and only the step-response regime of Section 2.3 applies.

2.5 Extension to a Maxwell visco-elastic asthenosphere

2.5.1 Constitutive model and the correspondence principle

The Maxwell relaxation time $\tau_R := \eta/\mu_m$ (where μ_m is the mantle shear modulus) separates two regimes: on timescales $t \ll \tau_R$ the asthenosphere responds elastically; on timescales $t \gg \tau_R$ it flows viscously (Bürgmann & Dresen 2008). We model this by replacing the Newtonian half-space with a Maxwell body: a Hookean spring and a Newtonian dashpot in series. The 1D constitutive relation for deviatoric stress τ_{ij} and strain ε is:

$$\dot{\varepsilon} = \frac{\dot{\tau}}{2\mu_m} + \frac{\tau}{2\eta}. \tag{25}$$

To adapt the Stokes kernel (3) we employ the elastic–viscoelastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010): in the Laplace domain, static moduli are replaced by their complex, frequency-dependent counterparts. Applying the Laplace transform $\tilde{f}(s) = \mathcal{L}\{f(t)\}$ to

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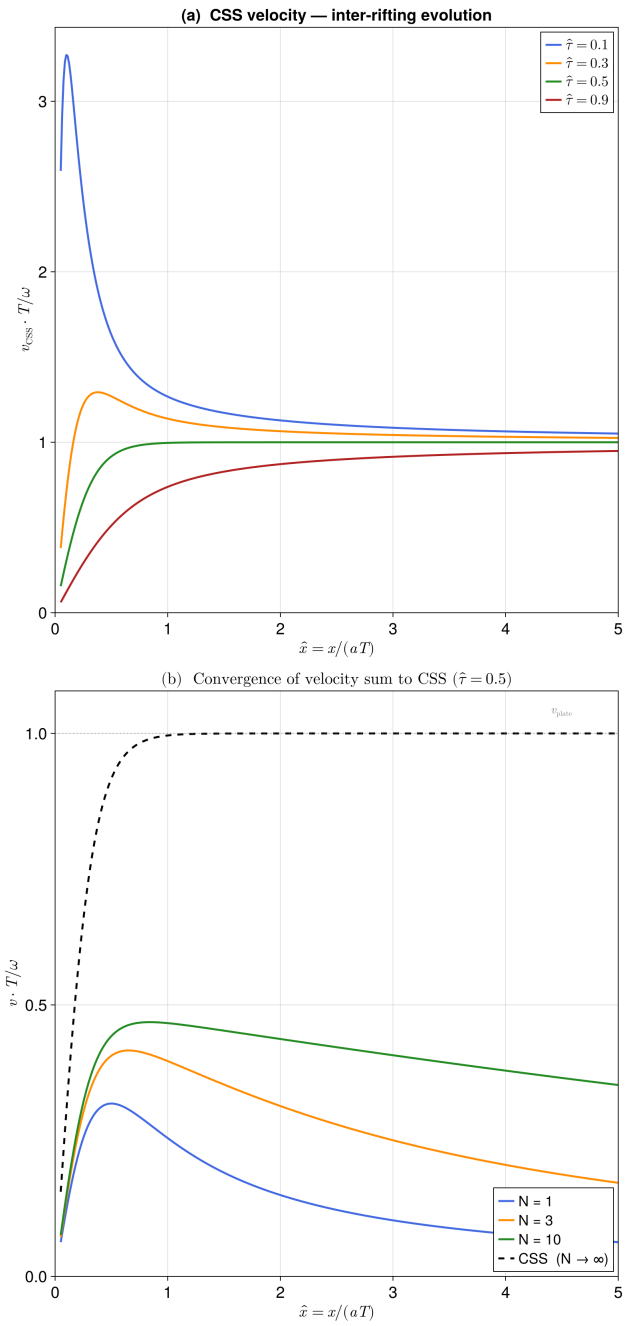


Figure 2. Dimensionless CSS velocity and convergence from finite rifting histories. All quantities are normalised by b and T; distances by aT. **(a)** CSS velocity $v_{\rm CSS} \cdot T/b$ versus $\hat{x} = x/(aT)$ at four inter-rifting phases $\hat{\tau} = \tau/T$. The velocity peak width is locked at aT regardless of phase; all profiles converge to $v_{\rm plate} = b/T$ (dotted line) in the far field. **(b)** Velocity at $\hat{\tau} = 0.5$ for $N = 1$, 3, and 10 dike events (solid curves) compared with the CSS limit (dashed). The $N = 1$ profile does not recover the far-field plate velocity; convergence to the CSS requires $\sim$10 cycles.

(25) with zero initial conditions:

$$s\tilde{\varepsilon} = \tilde{\tau}\left(\frac{s}{2\mu_m} + \frac{1}{2\eta}\right). \tag{26}$$

Solving for stress gives $\tilde{\tau} = 2\tilde{\mu}(s)\tilde{\varepsilon}$, where the effective frequency-dependent shear modulus of the Maxwell body is:

$$\tilde{\mu}(s) = \frac{\mu_m \eta\, s}{\mu_m + \eta s} = \mu_m \frac{\tau_R s}{1 + \tau_R s}. \tag{27}$$

2.5.2 *The visco-elastic Stokes kernel*

For a purely elastic half-space the basal traction relates to surface displacement through a kernel proportional to μ_m:

$$\sigma_{xz}^{\rm BOT,\ el}(x) = \int_{-\infty}^{+\infty} \frac{2\mu_m}{\pi(x - x')^2}\, u(x')\, dx'. \tag{28}$$

Applying the correspondence principle (replacing μ_m with $\tilde{\mu}(s)$ in the Laplace domain) and factoring out $\eta = \mu_m \tau_R$:

$$(1 + \tau_R s)\, \tilde{\sigma}_{xz}^{\rm BOT}(x, s) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x - x')^2} \left[s\tilde{u}(x', s)\right] dx'. \tag{29}$$

Inverting to the time domain (multiplication by s in the Laplace domain corresponds to $\partial/\partial t$) yields the visco-elastic Stokes kernel:

$$\sigma_{xz}^{\rm BOT} + \tau_R \frac{\partial \sigma_{xz}^{\rm BOT}}{\partial t} = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x - x')^2} \frac{\partial u(x', t)}{\partial t}\, dx'. \tag{30}$$

In the limit $\tau_R \to 0$, (30) reduces to the purely viscous Stokes kernel (3).

2.5.3 *Governing equation in Fourier space*

The equilibrium equation (2) expresses the basal traction in terms of the elastic crust displacement:

$$\sigma_{xz}^{\rm BOT} = -\frac{EH}{1 - \nu^2} \frac{\partial^2 u}{\partial x^2}. \tag{31}$$

Substituting (31) and its time derivative into (30), then applying the spatial Fourier transform using (5):

$$\frac{EH}{1 - \nu^2} k^2 \hat{u} + \tau_R \frac{EH}{1 - \nu^2} k^2 \frac{\partial \hat{u}}{\partial t} = -2\eta |k| \frac{\partial \hat{u}}{\partial t}. \tag{32}$$

Dividing by 2η, using (7), and collecting $\partial\hat{u}/\partial t$ terms gives the visco-elastic analogue of (8):

$$\frac{\partial \hat{u}}{\partial t} = -\left(\frac{a|k|}{1 + \tau_R a|k|}\right) \hat{u}. \tag{33}$$

The denominator $1 + \tau_R a|k|$ acts as a low-pass filter on the relaxation rate, slowing the decay of high-wavenumber (near-field) features relative to the purely viscous case.

2.5.4 *Step response and asymptotic behavior*

Equation (33) is separable in k, yielding a decaying exponential for each wavenumber:

$$\hat{u}(k, t) = \hat{u}(k, 0) \exp\left(-\frac{a|k|}{1 + \tau_R a|k|}\, t\right). \tag{34}$$

Applying the Heaviside initial condition (14):

$$\hat{u}(k, t) = b\pi\delta(k) + \frac{ib}{k} \exp\left(-\frac{a|k|}{1 + \tau_R a|k|}\, t\right). \tag{35}$$

Taking the inverse Fourier transform and following the same parity argument as in Section 2.3, the Maxwell step displacement requires numerical integration:

$$u^{\mathrm{MAX}}(x,t) = \frac{b}{2} - \frac{b}{\pi}\int_0^{+\infty} \frac{\sin(kx)}{k}\exp\!\left(-\frac{akt}{1+\tau_R ak}\right) dk, \tag{36}$$

since the rational exponent prevents the closed-form Dirichlet evaluation of Section 2.3. The surface velocity follows by differentiating (36) with respect to t:

$$v^{\mathrm{MAX}}(x,t) = \frac{ba}{\pi}\int_0^{+\infty} \frac{\sin(kx)}{1+\tau_R ak}\exp\!\left(-\frac{akt}{1+\tau_R ak}\right) dk. \tag{37}$$

Both integrals are evaluated numerically using adaptive Gauss-Kronrod quadrature (Laurie 1997), with integration breakpoints placed at the successive zeros of $\sin(kx)$ (i.e. $k = n\pi/x$, $n = 1, 2, \ldots$) to partition the oscillatory integrand into well-behaved sub-intervals. The upper truncation wavenumber is set adaptively to ensure the exponential factor is negligible at the integration cutoff.

The exponent in (36) reveals two physically distinct regimes. In the long-wavelength limit ($k \to 0$ or $t \to \infty$), $1 + \tau_R ak \approx 1$ and the exponential reduces to $\exp(-akt)$, recovering the purely viscous Dirichlet integral of Section 2.3. The elastic memory of the asthenosphere fades at large distances and late times, and the model converges to the Newtonian viscous limit. In the short-wavelength limit ($k \to \infty$ or $t \to 0$), $1 + \tau_R ak \approx \tau_R ak$ and the exponent saturates:

$$\lim_{k\to\infty}\left(-\frac{akt}{1+\tau_R ak}\right) = -\frac{t}{\tau_R}. \tag{38}$$

At early times $t \ll \tau_R$, the exponential approaches unity and no near-field deformation occurs outside the rift interface. For $t \sim \tau_R$, near-field deformation decays exponentially at the fixed rate $1/\tau_R$, representing pure Maxwell relaxation of the mantle independent of the elastic crust. Together, (37) and (38) demonstrate that the Maxwell model introduces a short-time, near-field stiffening relative to the viscous model: both converge to the same Cyclo-Steady State at $t \gg \tau_R$ (Section 2.6).

2.6 Maxwell Cyclo-Steady State

2.6.1 *Infinite sum and Fourier periodicity condition*

The Maxwell CSS is the superposition of step-response velocities from all preceding intrusions, by the same argument as in Section 2.4:

$$v^{\mathrm{MAX}}_{\mathrm{CSS}}(x,\tau) = \sum_{j=0}^{\infty} v^{\mathrm{STEP,MAX}}(x,\, \tau + jT). \tag{39}$$

Writing $v^{\mathrm{STEP,MAX}}$ in Fourier space (Section 2.5) as $\frac{2ba}{\pi}\int_0^\infty \sin(kx)\, e^{-\alpha(k)(\tau+jT)}\, dk$, with the effective Maxwell decay rate

$$\alpha(k) := \frac{ak}{1+\tau_R ak}, \tag{40}$$

each past intrusion contributes a factor $e^{-\alpha(k)jT}$. The series converges for all $k > 0$ and the geometric sum $\sum_{j=0}^{\infty} e^{-\alpha(k)jT} = (1 - e^{-\alpha(k)T})^{-1}$ can be performed inside the Fourier integral. The same result follows directly from the requirement that each Fourier mode satisfy the periodicity condition $\hat{u}(k, \tau + T) = \hat{u}(k, \tau) + ib/k$, which admits the unique solution

$$\hat{u}(k,\tau) = \frac{ib/k}{1 - e^{-\alpha(k)T}}\, e^{-\alpha(k)\tau}, \qquad \tau \in [0, T). \tag{41}$$

2.6.2 *Physical-space representation, conservation, and limits*

Inverting (41) gives the Maxwell CSS velocity and displacement:

$$v^{\mathrm{MAX}}_{\mathrm{CSS}}(x,\tau) = \frac{2ba}{\pi}\int_0^{\infty} \frac{\sin(kx)}{1+\tau_R ak}\,\frac{e^{-\alpha(k)\tau}}{1 - e^{-\alpha(k)T}}\, dk, \tag{42}$$

$$u^{\mathrm{MAX}}_{\mathrm{CSS}}(x,\tau) = \frac{2b}{\pi}\int_0^{\infty} \frac{\sin(kx)}{k}\,\frac{1 - e^{-\alpha(k)\tau}}{1 - e^{-\alpha(k)T}}\, dk. \tag{43}$$

Integrating the velocity over one full period, using $\int_0^T e^{-\alpha\tau}/(1 - e^{-\alpha T})\, d\tau = 1/\alpha$ and the algebraic identity $a/[(1 + \tau_R ak)\,\alpha(k)] = 1/k$, gives

$$\int_0^T v^{\mathrm{MAX}}_{\mathrm{CSS}}(x,\tau)\, d\tau = \frac{2b}{\pi}\int_0^{\infty} \frac{\sin(kx)}{k}\, dk = b, \tag{44}$$

where the last step is the Dirichlet integral. The full rift half-opening is recovered every cycle, independent of τ_R. In the limit $\tau_R \to 0$, $\alpha(k) \to ak$ and $(1 + \tau_R ak)^{-1} \to 1$, and (42) reduces to the viscous CSS Fourier integral whose closed form is the Digamma series of Section 2.4. For $\tau_R > 0$, the saturating denominator $1+\tau_R ak$ prevents this reduction and (42) is evaluated numerically.

2.7 Dimensionless form and model behavior

The viscous and Maxwell step responses each admit a natural non-dimensionalization that reveals the governing control parameters and delineates the physically distinct relaxation regimes.

2.7.1 *Viscous case: a universal single-parameter curve*

Introducing the dimensionless distance $\hat{x} = x/(at)$ (the spatial coordinate measured in units of the instantaneous diffusion front at), the step displacement (17) collapses to

$$\frac{u^{\mathrm{STEP}}}{b} = \frac{1}{\pi}\arctan\!\left(\frac{1}{\hat{x}}\right). \tag{45}$$

Equation (45) is a universal curve: every viscous step response, regardless of the values of E, H, η, or b, maps onto this single function of $\hat{x}$. Near the rift axis or at late times ($\hat{x} \ll 1$), the displacement saturates at $u/b \to 1/2$; far from the rift or at early times ($\hat{x} \gg 1$), $u/b \approx 1/(\pi\hat{x}) \to 0$. The corresponding velocity profile is an equally compact Lorentzian,

$$\frac{at\, v^{\mathrm{STEP}}}{b} = \frac{1}{\pi}\,\frac{\hat{x}}{1+\hat{x}^2}, \tag{46}$$

with a peak located precisely at the diffusion front ($\hat{x} = 1$, i.e. $x = at$), an amplitude that decays as $b/(at)$, and a width that broadens linearly in time. The single parameter a thus encodes the entire spatio-temporal evolution of the post-diking transient.

2.7.2 *Maxwell case: two dimensionless groups and three regimes*

The Maxwell step solution introduces a second dimensionless group. For positive wavenumbers $k > 0$, writing the Fourier-space exponent (34) as

$$-\frac{akt}{1+\tau_R ak} = -\frac{\hat{t}\,\kappa}{1+\kappa}, \tag{47}$$

where $\hat{t} = t/\tau_R$ is the Maxwell number and $\kappa = \tau_R ak$ is the wavenumber scaled by the Maxwell length

$$\ell_M := a\tau_R = \frac{EH}{2\mu_m(1-\nu^2)}, \tag{48}$$

three physically distinct relaxation regimes emerge. (i) *Elastic rebound* ($\kappa \gg 1$, i.e. $x \ll \ell_M$): the exponent in (47) saturates at $-\hat{t}$, independent of wavenumber, so near-field deformation decays at the uniform rate $1/\tau_R$, pure Maxwell relaxation of the mantle spring, independent of the elastic crust. (ii) *Viscous diffusion* ($\hat{t} \gg 1$ or $x \gg \ell_M$, $\kappa \ll 1$): the denominator $1+\kappa \approx 1$ and the Maxwell solution converges pointwise to the purely viscous step response of Section 2.3. (iii) *Crossover* ($\hat{t} \sim 1$): both timescales are simultaneously active; near-field deformation lags behind the viscous prediction (a short-time stiffening) before relaxing toward the common Cyclo-Steady State at $t \gg \tau_R$. The Maxwell length $\ell_M = EH/[2\mu_m(1-\nu^2)]$ is a *purely elastic* quantity, independent of viscosity: it measures the ratio of lithospheric to mantle elastic stiffness times the plate thickness, and sets the spatial boundary between the two asymptotic regimes.

Figure 3 illustrates both regimes for the step response, with axes normalised by $\tau_{R,\mathrm{ref}}$ and $\ell_M = a\tau_{R,\mathrm{ref}}$. Panel (a) shows the rift-perpendicular velocity at the fixed snapshot time $t = \tau_{R,\mathrm{ref}}$ for four values of the physical relaxation time $\tau_R/\tau_{R,\mathrm{ref}}$: increasing τ_R progressively suppresses near-field velocity for $x \lesssim \ell_M$, leaving the far-field Lorentzian unchanged. Panel (b) traces the cumulative displacement at $x = \ell_M$ through the crossover: Maxwell models lag the viscous prediction for $t \lesssim \tau_R$ before converging to the common asymptote $b/2$. These two panels define an observational criterion. Post-diking geodetic records collected within a few τ_R of the rifting event are in the elastic rebound regime; the purely viscous step response underestimates near-field displacement and overestimates the deformation rate at $x < \ell_M$ in this window. Once $t \gg \tau_R$ the elastic spring has fully relaxed and the simpler viscous model is an accurate substitute.

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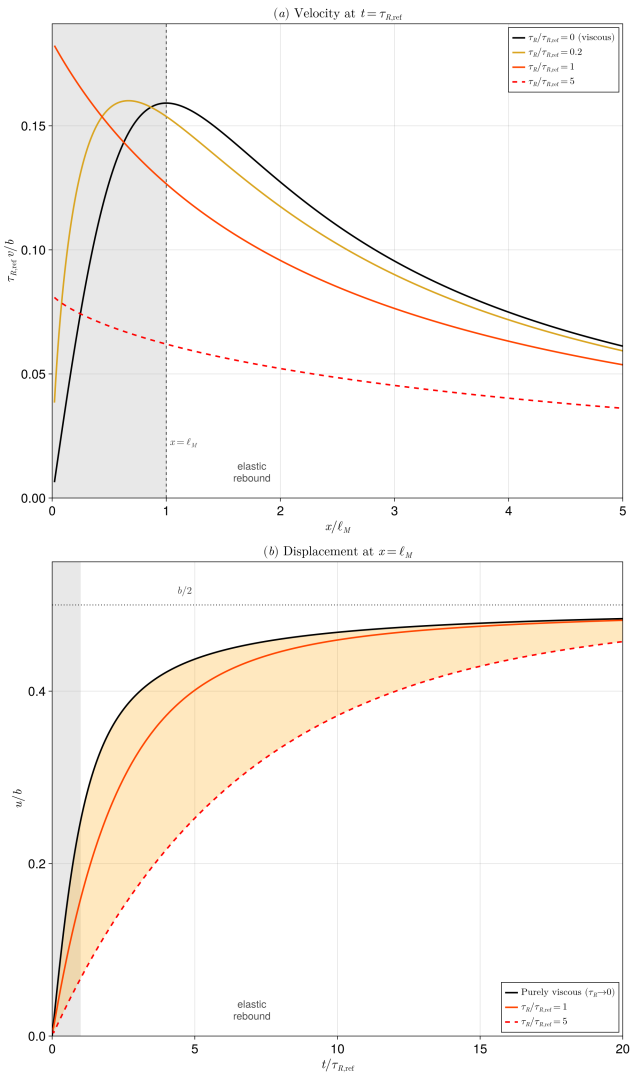


Figure 3. Maxwell visco-elastic two-regime behavior. Distances are normalised by the Maxwell length $\ell_M = a\tau_{R,\mathrm{ref}}$, times by $\tau_{R,\mathrm{ref}}$, velocities by ba/π, and displacements by b. Grey shading marks the elastic-rebound zone ($x < \ell_M$ in panel **a**; $t < \tau_R$ in panel **b**). **(a)** Velocity at $t = \tau_{R,\mathrm{ref}}$ for four values of $\tau_R/\tau_{R,\mathrm{ref}}$ (0, 0.2, 1, 5). The black curve is the purely viscous Lorentzian, peaking at $x = \ell_M$ (dashed line). Increasing τ_R suppresses near-field velocity for $x \lesssim \ell_M$; at $\tau_R/\tau_{R,\mathrm{ref}} = 5$ (dashed red) the near-field amplitude is reduced by more than 60%. **(b)** Cumulative displacement at $x = \ell_M$ for the viscous model (black) and two Maxwell cases ($\tau_R/\tau_{R,\mathrm{ref}} = 1$, orange; $= 5$, dashed red). Orange shading marks the displacement deficit relative to viscous. Both Maxwell models converge to the common asymptote $b/2$ (dotted) after $t \gg \tau_R$.

2.7.3 *Cyclo-Steady State: the deformation-width scaling*

The CSS introduces a different natural length scale. In the viscous CSS (22), the Digamma argument is $\tau/T + ix/(aT)$, so the deformation width scales as aT rather than at. Defining $\hat{x}_{\mathrm{CSS}} = x/(aT)$ and $\hat{\tau} = \tau/T \in [0,1)$, the CSS velocity collapses to

$$\frac{T\,v_{\mathrm{CSS}}}{b} = \frac{2}{\pi}\,\mathrm{Im}\{\psi(\hat{\tau} + i\hat{x}_{\mathrm{CSS}})\}\,, \tag{49}$$

a universal function of $(\hat{\tau},\, \hat{x}_{\mathrm{CSS}})$ alone. Unlike the step response, the CSS velocity profile does not broaden with time: its width is locked at aT by the periodicity of the forcing. For the Maxwell CSS (42), the ratio τ_R/T enters as a third dimensionless group. When $\tau_R/T \ll 1$ the Maxwell CSS converges pointwise to (49); when $\tau_R/T \sim 1$, near-field stiffening persists into the steady state, narrowing the velocity peak relative to the purely viscous prediction (Figure 3).

3 RESULTS: APPLICATIONS TO MAGMATIC RIFT SYSTEMS

Post-diking geodetic records from two contrasting rift environments test the models developed in Section 2 and constrain sub-lithospheric viscosity across distinct tectonic settings. We apply the viscous step response (Section 2.3) to a rift-perpendicular differenced displacement profile from the 1975–1984 Krafla rifting sequence in NE Iceland (Foulger et al. 1992) and to a 25-year baseline time series combined with a rift-perpendicular velocity profile from the 1978 Asal–Ghoubbet episode in the Afar Triangle (Vigny et al. 2007). Both datasets fall within the single-event step-response regime: at 8–11 yr post-rifting, Krafla has experienced less than one recurrence interval, and the Asal record ends before a subsequent major rifting episode. Together, the two sites span a factor-of-five contrast in best-fit characteristic velocity a and probe the rheological contrast between a plume-fed mid-ocean rift and an early-stage continental rift.

At each site, model parameters are estimated by minimizing the chi-squared misfit

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{u_i^{\mathrm{obs}} - u_i^{\mathrm{pred}}}{\sigma_i} \right)^2 , \tag{50}$$

where u_i^{obs} and u_i^{pred} are the observed and predicted displacement (or baseline change) at data point i, σ_i is the 1σ measurement uncertainty, and N is the number of observations. Goodness of fit is quantified through the reduced statistic $\chi^2_r = \chi^2/(N-p)$, where p is the number of free parameters; $\chi^2_r \approx 1$ indicates that the model reproduces the data within the measurement noise.

3.1 Krafla rift, Iceland

The 1984 Krafla rifting episode in NE Iceland provides one of the best-documented post-diking geodetic records available. Foulger et al. (1992) measured horizontal surface displacements at 20 stations across the rift zone at two epochs 8 and 11 years after the 1975–1984 rifting sequence. We fit the differenced displacement field $\Delta u = u(\tau_f) - u(\tau_i)$, with $\tau_i = 8$ yr and $\tau_f = 11$ yr, using three forward models: (i) the diffusion model of Foulger et al. (1992), with free parameters (κ, b); (ii) the viscous step response (Section 2.3), with free parameters (a, b); and (iii) the Maxwell step response (Section 2.5), with free parameters (a, b, τ_R). For each model, parameters are estimated by minimizing equation (50) over a Latin Hypercube sample of 10^4 points in log-uniform parameter space, followed by bounded Nelder-Mead refinement. At 8–11 yr post-rifting the system has completed at most a fraction of one recurrence interval ($T \gtrsim 100$ yr in Iceland), placing the observations firmly in the transient step-response regime; the CSS representations of Sections 2.4 and 2.6 are therefore not applied here.

All three models fit the observed displacement profile within the measurement noise (Figure 4). The viscous step response ($a = 5.13\times10^{-5}$ m s^{-1}, $b = 1.14$ m; $\chi^2_r = 0.72$) and the Foulger diffusion model ($\kappa = 0.62$ m^2 s^{-1}, $b = 0.81$ m; $\chi^2_r = 0.90$) are both consistent with the data within the noise level. Since the two models are non-nested, their comparison requires information-theoretic criteria: the Akaike Information Criterion (AIC; Akaike 1974) gives $\Delta\mathrm{AIC} = -3.3$ in favour of the viscous step response, constituting positive evidence against the Foulger model ($|\Delta\mathrm{AIC}| > 2$). Leave-one-out cross-validation (Stone 1974) confirms this ranking: the mean squared normalised prediction error is 0.84 for the step response versus 1.02 for the Foulger model. Adding the Maxwell relaxation time as a third free parameter yields $a = 6.61\times10^{-5}$ m s^{-1}, $b = 1.15$ m, $\tau_R = 1.48$ yr ($\chi^2_r = 0.77$); however, three independent criteria indicate that τ_R is not constrained by these data: the likelihood-ratio test gives $\Delta\chi^2 = 2.53 < 3.84$ for one additional degree of freedom (Wilks 1938), the AIC penalises the extra parameter, and the leave-one-out prediction error for the Maxwell model (1.08) is the worst of the three, a signature of overfitting. The value $\tau_R = 1.48$ yr should therefore be treated as an upper bound on the Maxwell relaxation time compatible with these observations, not as a constrained estimate. Using the viscous best-fit a in (7) yields $\eta = EH/[2a(1-\nu^2)] \approx 4.2\times10^{18}$ Pa s for canonical Icelandic lithospheric parameters ($H = 10$ km, $E = 40$ GPa, $\nu = 0.25$), consistent with post-glacial rebound estimates (Sigmundsson et al. 2010) and below the upper bound $\eta < 10^{19}$ Pa s placed by Sigmundsson (1991) for a 10-km elastic layer at Krafla. The joint χ^2/N landscape in the (a, b) plane (Figure 5) confirms that both parameters are well-resolved by these data.

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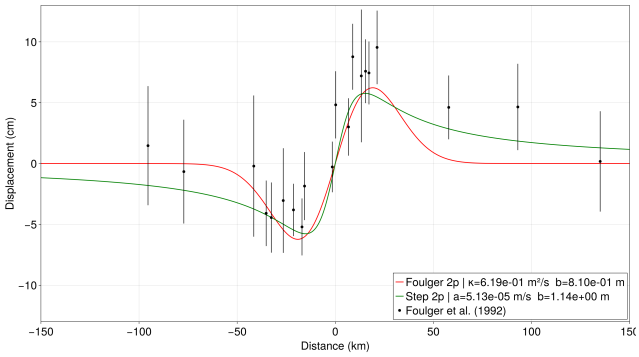


Figure 4. Best-fit differenced displacement profiles for the Krafla post-diking dataset of Foulger et al. (1992). Observed rift-perpendicular displacements between $\tau_i = 8$ yr and $\tau_f = 11$ yr post-rifting are shown as filled circles with 1σ error bars (cm). Curves show the best-fit Foulger diffusion model (red), viscous step response (green), and Maxwell visco-elastic response (orange); see text for parameter values and goodness-of-fit statistics.

3.2 Asal rift, Djibouti

The November 1978 Asal–Ghoubbet rifting episode in the Afar Triangle opened approximately 2 m of rift-perpendicular extension over several days (Le Dain et al. 1979; Ruegg et al. 1984), launching one of the longest continuously monitored post-diking transient records outside Iceland. Vigny et al. (2007) tracked the EP00–DF00 baseline (a rift-perpendicular geodetic tie crossing the Asal rift) from 1979 to 2003, recording $\approx$ 490 mm of cumulative lengthening, with rapid early growth decelerating sharply after the mid-1980s. We digitize all 33 data points from Figure 3 of Vigny et al. (2007), combining trilateration campaigns (1979–1987) and GPS surveys (1991–2003), and fit the two forward models following the approach of Section 3.1. The

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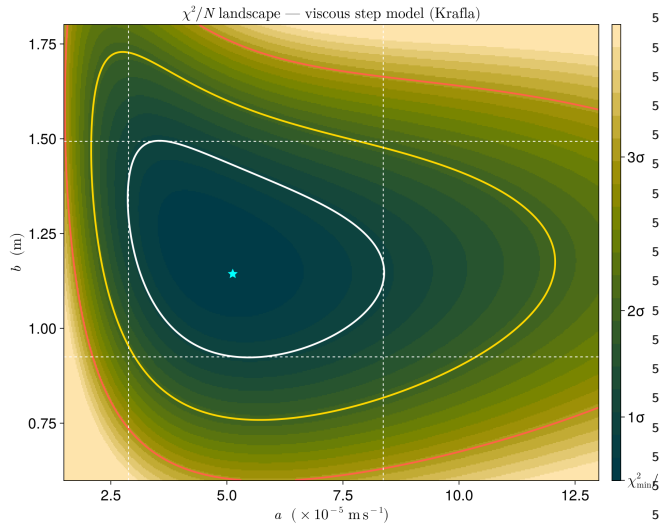


Figure 5. Reduced chi-squared (χ^2/N) landscape in the (a, b) parameter plane for the Krafla viscous step fit. Colour encodes χ^2/N relative to the minimum; white, yellow-orange, and red contours mark the 68.3% (1σ), 95.4% (2σ), and 99.7% (3σ) joint confidence regions. The cyan star marks the best-fit $(a^*, b^*) = (5.13 \times 10^{-5}$ m s^{-1}, 1.14 m); dashed lines project the marginal 1σ intervals onto each axis.

predicted baseline change is evaluated as $\Delta L = 2u(x_0, t)$ at $x_0 = 2000$ m, half the EP00–DF00 rift-perpendicular separation, with the rifting onset fixed at $t_0 = 1978.85$ yr (Vigny et al. 2007). Because the Vigny et al. time series is referenced to the first measurement epoch ($t_{\rm ref} \approx 1.1$ yr after the rifting event) rather than to the rifting onset, we subtract from each model curve its own prediction at $t_{\rm ref}$, anchoring $\Delta L = 0$ at the reference epoch without introducing a free parameter.

Neither model fits the Asal time series decisively better than the other (Figure 6). The best-fit Foulger diffusion parameters are $\kappa = 1.04 \times 10^{-2}$ m^2 s^{-1} and $U_0 = 0.44$ m ($\chi^2_r = 1.47$); the viscous step response yields $a = 9.5 \times 10^{-6}$ m s^{-1} and $b = 0.71$ m ($\chi^2_r = 1.45$). The AIC difference between the two models is $\Delta{\rm AIC} = -0.84$ in favour of the viscous step response, well below the threshold of 2 that would constitute positive evidence for either model (Burnham & Anderson 2002); leave-one-out cross-validation gives normalised mean squared prediction errors of 1.64 and 1.52, respectively, confirming this ranking. The published Foulger fit of Vigny et al. (2007) (
kappa = 0.015 m^2 s^{-1}, $U_0 = 0.4$ m) performs substantially worse on the full 33-point dataset ($\chi^2_r = 4.75$), consistent with those parameters being rounded estimates from an informal fit rather than the result of a weighted χ^2 minimisation. This result contrasts with Krafla (Section 3.1), where the viscous step response was decisively favoured; at Asal, the data alone cannot discriminate between the two rheological models. The χ^2/N landscape for the viscous step fit (Figure 8) shows well-resolved parameters with a narrow confidence ellipse elongated along the b axis.

Physical constraints from independent field observations break the statistical degeneracy in favour of the viscous step response. The best-fit Foulger half-opening $U_0 = 0.44$ m implies a total dike opening of $2U_0 = 0.88$ m, roughly half the ≈ 1.9 m measured across the rift by trilateration immediately after the 1978 event (Ruegg et al. 1979). The viscous step response yields $b = 0.71$ m, corresponding to a total opening of $2b = 1.43$ m, closer to the field measurement by a factor of two. The same discrepancy appears in the early deformation rate: the viscous model predicts a separation rate of ≈ 66 mm yr^{-1} at $t \approx 1.1$ yr, consistent with the ≈ 53 mm yr^{-1} measured by trilateration shortly after the event (Ruegg & Kasser 1987), whereas the Foulger model predicts only ≈ 47 mm yr^{-1}. This difference arises because the erfc displacement profile requires the diffusion front to propagate outward before the station at $x_0 = 2$ km responds, suppressing early deformation; the arctan profile of the viscous step response distributes strain outward immediately from $t = 0$. The best-fit parameter $a = 9.5 \times 10^{-6}$ m s^{-1} implies a sub-lithospheric viscosity of $\eta \approx 2.2 \times 10^{19}$ Pa s for Afar lithospheric parameters ($H = 10$ km, $E = 40$ GPa, $\nu = 0.25$), in the range of estimates from other post-diking transient analyses in the East African Rift (Nooner et al. 2009).

Vigny et al. (2007) also measured rift-perpendicular GPS velocities along a transect crossing the rift at a mean epoch of ≈ 19 yr after the 1978 event. We predict the velocity profile from the time-series best-fit parameters, placing the model centre at $x = +1$ km northeast of the topographic rift axis to coincide with the Petit Rift (Vigny et al. 2007). Both models reproduce the near-rift velocity gradient (Figure 7). The dominant misfit is the ≈ 16 mm yr^{-1} Somalia–Danakil background plate motion (Saria et al. 2014; Stamps et al. 2021), which produces a pronounced SW–NE velocity asymmetry absent from our post-rift transient models. After removing a best-fit uniform background velocity ($V_{\rm plate} = 8$ mm yr^{-1}), the viscous step model provides a better spatial fit ($\chi^2_r = 18$) than the Foulger model ($\chi^2_r = 33$): the erfc profile decays more rapidly with distance than the Lorentzian step profile, underpredicting velocities at stations several kilometres from the rift axis. Neither model achieves $\chi^2_r \sim 1$; the elevated residuals reflect the asymmetric far-field signal from Somalia–Danakil plate motion that a single symmetric transient and a uniform velocity offset cannot fully capture.

4 DISCUSSION

4.1 A boundary-condition-free description of inter-rifting deformation

The CSS framework closes the kinematic gap identified in Section 1 without introducing a locking depth, a prescribed domain width, or an artificially imposed far-field velocity. Backslip models of inter-rifting deformation (Savage & Prescott 1978) and their dislocation equivalents (LaFemina et al. 2005; Konca et al. 2018) reproduce observed velocity gradients only at the cost of a free geometric parameter (the locking depth or creep-zone geometry) that carries no independent mechanical constraint in magmatic rifts where the rheological transition is gradational rather than sharp. Numerical visco-elastic simulations face an equivalent obstacle: in any finite domain, lateral boundary conditions must either force far-field velocities to zero, in contradiction with global plate motion models (Altamimi et al. 2016), or pre-

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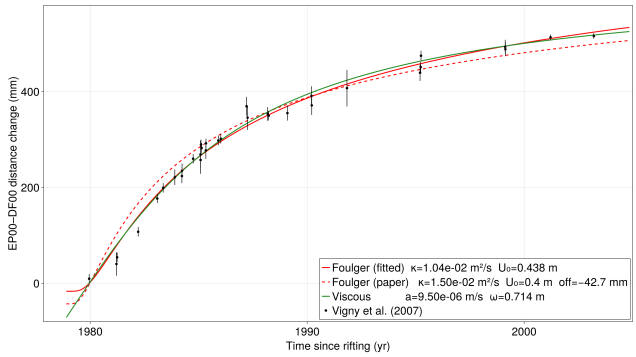


Figure 6. Best-fit forward models for the Asal post-diking time series of Vigny et al. (2007). The EP00–DF00 baseline change (mm) is shown as filled circles with 1σ error bars. Solid curves show the best-fit Foulger diffusion model (red) and viscous step response (green); the dashed red curve shows the published Foulger fit of Vigny et al. (2007). All curves are referenced to the first measurement epoch ($t_{\mathrm{ref}} \approx 1.1$ yr); see text for parameter values and goodness-of-fit statistics.

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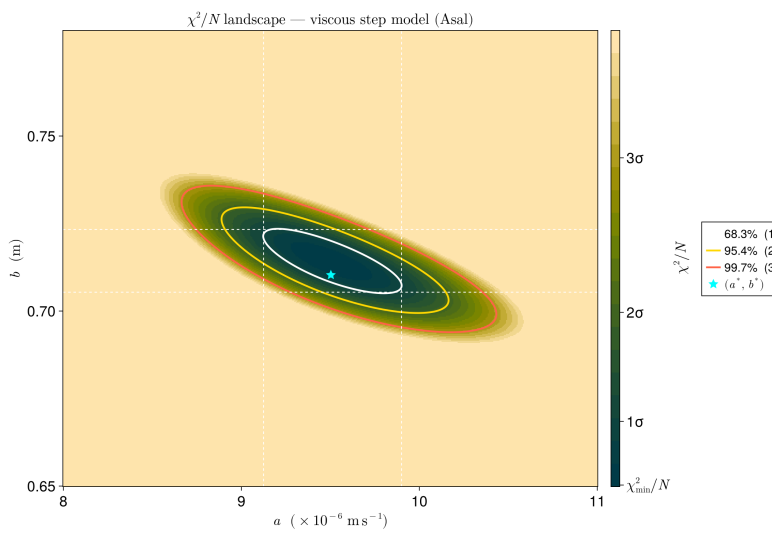


Figure 8. Reduced chi-squared (χ^2/N) landscape in the (a, b) parameter plane for the Asal viscous step fit. Contours and symbols as in Figure 5. The best-fit is (a^*, b^*) = (9.5×10^{-6} m s^{-1}, 0.71 m). The narrow, elongated confidence ellipse reflects the tight constraint on a imposed by the 25-yr baseline change time series.

scribe the divergence rate that the model is itself meant to predict (Hofton et al. 1995; Hofton & Foulger 1996).

The CSS requires neither. The deformation zone half-width $aT = EHT/[2\eta(1-\nu^2)]$ emerges directly from the balance between elastic restoring stresses in the plate and viscous drag at its base, parameterized entirely by the characteristic velocity a and the recurrence interval T. Far-field plate divergence is satisfied by construction through the conservation property of the Digamma representation (equation 24): no boundary condition is imposed, and no artificial velocity offset is needed to reconcile model predictions with far-field geodesy. Both free parameters are physically grounded: the dike half-opening b is measurable independently from field geology and geodetic surveys (Le Dain et al. 1979; Ruegg et al. 1984), and a directly constrains the ratio EH/η without requiring an independent estimate of viscous layer thickness. This structural simplicity distinguishes the CSS from existing kinematic and semi-analytical approaches to the inter-rifting problem.

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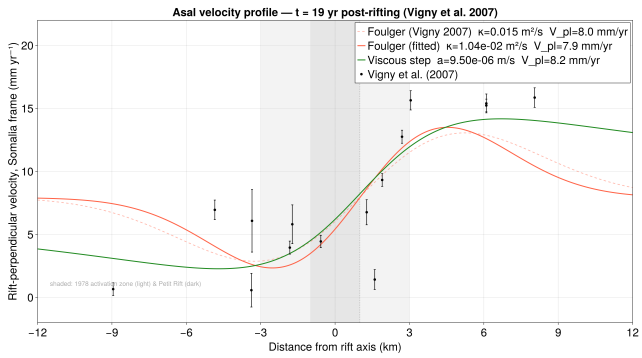


Figure 7. Forward-predicted rift-perpendicular velocity profiles at $t \approx 19$ yr post-rifting, compared with GPS observations of Vigny et al. (2007) (Somalia reference frame, mm yr^{-1}). Model parameters are fixed at the time-series best-fit values (Figure 6); a uniform plate-motion offset ($V_{\mathrm{plate}} = 8$ mm yr^{-1}) has been added to each curve. Shading marks the 1978 activated zone (±3 km, light) and the Petit Rift (±1 km, dark); the dashed line marks the model centre at $x = +1$ km.

4.2 Rheological constraints and deformation zone scaling

The single composite parameter $a = EH/[2\eta(1-\nu^2)]$ collapses the elastic–viscous trade-off onto a one-dimensional family of solutions in (E, H, η) space. Figure 9 makes this trade-off explicit: for fixed E, the iso-a contour traces a band across (H, η) space. Dark shading propagates the 1σ uncertainty on a from the χ^2 fit at fixed $E = 40$ GPa; light shading additionally propagates E over [20, 80] GPa (combined 1σ envelope). The iso-a curves are nearly parallel in this representation, so the dominant uncertainty is in η rather than in H. Independent seismological constraints on the seismogenic layer thickness ($H \approx 5$–15 km in both Iceland and the Afar; Stevens et al. 2021; Plasman et al. 2017; Weinstein et al. 2017 break the remaining ambiguity and allow us to translate the geodetically constrained a into a viscosity estimate for each site (Figure 9).

Adopting $H = 10$ km and $E = 40$ GPa, the Krafla best-fit $a = 5.13 \times 10^{-5}$ m s^{-1} implies $\eta \approx 4.2 \times 10^{18}$ Pa s. Two independent sources of uncertainty affect this estimate. The 1σ confidence interval on a from the χ^2 fit (Section 3) yields $\eta \in [2.6, 7.4] \times 10^{18}$ Pa s for fixed $E = 40$ GPa, while propagating E over [20, 80] GPa at the best-fit a gives $\eta \in [2.1, 8.3] \times 10^{18}$ Pa s. Because $\eta = EH/[2a(1-\nu^2)]$, the two uncertainties are additive in logarithmic space; combining them in quadrature ($\sigma_{\log_{10} E} = 0.30$, $\sigma_{\log_{10} a} = 0.23$) yields a combined 1σ range of $\eta \in [1.7, 10.0] \times 10^{18}$ Pa s (Figure 9, left). This range lies entirely below or at the upper bound $\eta < 10^{19}$ Pa s placed by Sigmundsson (1991) for a 10-km elastic layer at Krafla, and is consistent with

post-glacial rebound estimates across Iceland ($\eta \sim 10^{18}$–10^{19} Pa s; Sigmundsson et al. 2010) and with forward visco-elastic modelling of the same post-diking transient (Hofton & Foulger 1996). The low viscosity beneath Iceland reflects anomalously high temperatures in the sub-lithospheric mantle maintained by the Iceland plume: partial melt in the lower crust and uppermost mantle reduces the effective viscosity by one to two orders of magnitude relative to standard continental lithosphere (Holtzman 2016).

The Asal best-fit $a = 9.5 \times 10^{-6}$ m s^{-1} implies $\eta \approx 2.2 \times 10^{19}$ Pa s at $H = 10$ km and $E = 40$ GPa, with an E-propagated range of $\eta \in [1.1, 4.5] \times 10^{19}$ Pa s, approximately five times higher than the Krafla estimate. This is higher than viscosity estimates for the Dabbahu segment of the same rift system ($\eta \approx 5 \times 10^{18}$ Pa s; Nooner et al. 2009; $\eta \sim 10^{18}$ Pa s, $H = 21$ km; Hamling et al., 2014 [CITE NEEDED]), reflecting either genuine segment-to-segment variation in sub-lithospheric temperature or contributions from shallow magmatic sources that the purely viscous step model cannot distinguish from mantle relaxation. Smittarello et al.
(2016) [CITE NEEDED] argue, from 38 years of geodetic observations of the Asal–Ghoubbet rift, that magma supply rather than viscoelastic relaxation controls the post-diking velocity field at this site, a degeneracy our single-parameter model cannot resolve without independent constraints on the magmatic budget. The factor-of-five viscosity contrast between Krafla and Asal nevertheless remains physically coherent: both are magmatically active rifts, but Iceland's plume-driven heat flux maintains substantially higher sub-lithospheric temperatures, accelerating viscous relaxation relative to the East African setting.

Figure 10 translates the best-fit a values into predictions for the CSS deformation zone width aT, which controls how broadly inter-rifting velocities are distributed across a geodetic network. For a Krafla recurrence interval of $T \approx 200$ yr, consistent with the ~ 250 yr gap between the 1724–1729 and 1975–1984 eruptive crises, the CSS width reaches $aT \approx 324$ km, already exceeding the 230 km span of the Foulger et al. (1992) network (Figure 10), consistent with the broad velocity gradients observed across Iceland (LaFemina et al. 2005). For Asal the recurrence interval is poorly constrained; adopting $T \approx 500$ yr gives $aT \approx 150$ km, nearly ten times the 17 km aperture of the Vigny et al. (2007) GPS profile (Figure 10). In both cases the CSS width vastly exceeds the network aperture, which is precisely why inter-rifting velocity fields in Iceland and the Afar have been interpreted as steady plate divergence rather than as the superposition of slowly decaying visco-elastic transients.

4.3 Non-local diffusion versus the Fickian model

The preference for the viscous step response over the Foulger diffusion model is not a numerical accident: it reflects a structural difference in the governing equations. Foulger et al. (1992) model post-diking deformation as Fickian diffusion of stress, governed by $\partial u/\partial t = \kappa\, \partial^2 u/\partial x^2$, whose Fourier signature is $\partial \hat{u}/\partial t = -\kappa k^2 \hat{u}$. The quadratic wavenumber dependence concentrates decay at short wavelengths and generates an erfc displacement profile with a deformation front that propagates as $\sqrt{\kappa t}$ and leaves the far field exponentially unperturbed on timescales shorter

[htbp]

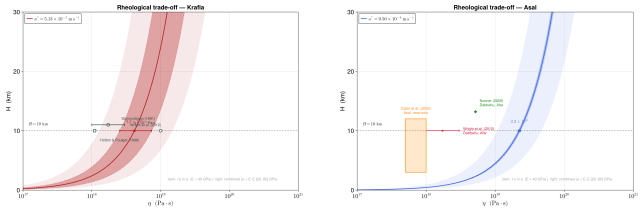


Figure 9. Rheological trade-off for Krafla (left, red) and Asal (right, blue). Each panel shows the iso-a contour in the (H, η) plane at the best-fit value. Dark shading propagates the 1σ uncertainty on a from the χ^2 fit at fixed $E = 40$ GPa; light shading additionally propagates E over [20, 80] GPa (combined 1σ envelope). The $H = 10$ km dashed line marks the reference elastic thickness; markers show published viscosity estimates (Krafla: open circle, Sigmundsson 1991; open square, Hofton & Foulger 1996; open circle with error bar, Wright et al. 2012; Asal: filled diamond, Nooner et al. 2009 for the Dabbahu segment of the same rift system; orange box, Cattin et al. 2005 — near-axis effective viscosity $\eta = 0.5$–1×10^{18} Pa s from a 2D thermo-mechanical FEM model of the first 6–8 yr post-rifting at Asal, with the seismogenic layer thickness $H = 3$–12 km from the same study). The Cattin et al. box lies well below the iso-a^* curve, consistent with the local near-axis viscosity being lower than the laterally averaged bulk estimate from the 25-yr geodetic record.

than x^2/κ. Our viscous model is governed instead by (8), $\partial \hat{u}/\partial t = -a|k|\hat{u}$, where the absolute value $|k|$ is the Fourier representation of the half-Laplacian $(-\partial^2/\partial x^2)^{1/2}$, a spatially *non-local* operator arising from the $1/(x - x')^2$ Stokes convolution kernel (3). Because the mantle transmits stress instantaneously through the incompressible Stokes flow, every point on the elastic plate feels the rifting event from the

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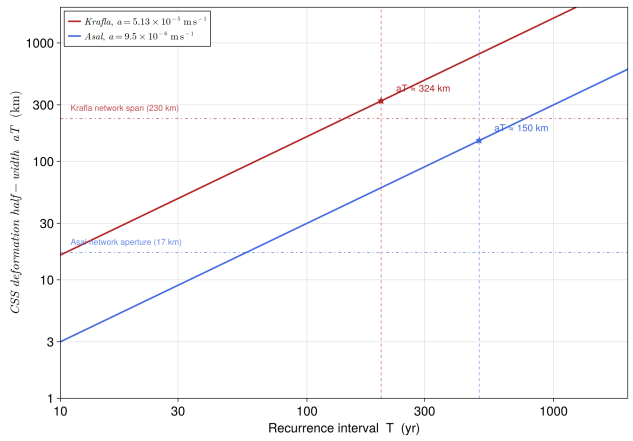


Figure 10. CSS deformation half-width aT as a function of recurrence interval T for Krafla (red, $a = 5.13 \times 10^{-5}$ m s^{-1}) and Asal (blue, $a = 9.5 \times 10^{-6}$ m s^{-1}). Vertical dashed lines mark estimated recurrence intervals; annotated points give the implied aT values. Horizontal dash-dot lines mark the total span of each geodetic network used in this study: 230 km for Krafla (Foulger et al. 1992, 1992; distance between outermost stations) and 17 km for Asal (Vigny et al., 2007; GPS profile extent). In both cases aT substantially exceeds the network aperture, explaining why inter-rifting deformation has been interpreted as steady plate divergence rather than a broad visco-elastic transient.

first instant; no diffusive front is required to reach it. The surface velocity is therefore a Lorentzian (46), with tails decaying as $\sim 1/x^2$ rather than the exponential or Gaussian decay of the Fickian model.

The observational consequence is directly visible in the Asal velocity transect (Figure 7). At stations several kilometres from the rift axis (beyond the zone where model differences are masked by near-field noise), the viscous step response fits the data better than the Foulger model ($\chi^2_r = 18$ vs 33, a factor-of-two improvement), because the Lorentzian distributes deformation more broadly than the erfc profile. The same asymmetry drives the profile misfit more clearly at Asal than at Krafla, where the differenced-displacement observable $\Delta u = u(\tau_f) - u(\tau_i)$ suppresses far-field signal regardless of the spatial functional form. The non-local character of the Stokes kernel thus has a concrete, testable imprint on the spatial pattern of inter-rifting deformation: networks sampling distances $x \gg aT/\pi$ are more sensitive to the tail behaviour and therefore better placed to discriminate between the two formulations.

The two formulations are not independent alternatives, however: the Foulger diffusion profile is mathematically embedded within the viscous solution. Bochner's subordination formula writes the half-Laplacian semigroup as a weighted average, over all internal diffusion times, of the ordinary Fickian heat semigroup (Metzler & Klafter 2000). Substituting the model's own initial condition into this average reproduces the closed-form viscous step response exactly. The Foulger model is therefore not simply the wrong physical model: it is the correct solution evaluated at a single diffusion timescale, rather than blended across all of them.

4.4 Robustness under stochastic recurrence

The CSS derivation assumes strictly periodic rifting: intrusions occur at fixed interval T and deliver a fixed half-opening b at each event. Neither assumption holds in nature. Magmatic rifting sequences are episodic and irregular; the $\sim$ 250-year gap between the Krafla 1724–1729 and 1975–1984 crises is a single estimate from a sparse historical record, and dike opening magnitudes vary substantially within and between sequences (Thordarson & Larsen 2007). The CSS remains a robust estimator of the long-term inter-rifting velocity field even under substantial cycle-to-cycle variability in timing and dike opening, as we demonstrate through stochastic ensembles.

We replaced the deterministic event sequence with stochastic ensembles in which each past rifting event draws its recurrence interval T_j and half-opening b_j independently from a Normal distribution with mean (T, b) and coefficient of variation $c_v = 0.30$, a scatter representative of documented variability in Icelandic and Afar rift chronologies. For each of 500 realisations we summed the step-response contributions from all past events and extracted the 10th–90th percentile band across the ensemble (Figure 11).

Four results stand out. First, the ensemble median coincides with the deterministic CSS in all four cases: linearity of the step-response sum and the conditions $\mathbb{E}[T_j] = T$ and $\mathbb{E}[b_j] = b$ guarantee that the ensemble average recovers the periodic solution exactly, regardless of the scatter level. Second, the spread differs sharply between the two sources of variability. Scatter in T (panel a) concentrates uncertainty in the transition zone near $\hat{x} \sim 1$: irregular timing shifts the velocity gradient left or right across realisations, smearing the transition zone while the far-field plateau $v \to b/T$ remains sharp. Scatter in b (panel b) acts differently: because the recurrence interval is fixed, events arrive at the same times in every realisation and the profile shape is identical to the deterministic CSS in all of them; only its overall amplitude varies. The effect is equivalent to sliding each realisation's curve up or down the y-axis by a random factor, so the band at each distance is a fixed fraction of the local CSS value and neither the gradient position nor the far-field plateau is affected. Third, the combined scatter (panel c) produces the widest band; at the deformation-width scale $\hat{x} = 1$, the 10–90th percentile range spans roughly one-third of the median CSS amplitude for $c_v = 0.30$, and the band narrows steadily toward the far field as the law-of-large-numbers averaging over many past events takes effect. This width confirms that the CSS is a robust estimator of the time-averaged inter-rifting velocity field even under substantial cycle-to-cycle variability. Fourth, imposing the plate-rate-conserving coupling $b_j = v_{\mathrm{plate}} T_j$ (panel d) causes the far-field band to collapse to essentially zero: because $b_j/T_j = v_{\mathrm{plate}}$ for every event in every realisation, the plate velocity is encoded exactly regardless of timing scatter. Residual spread in the transition zone ($|\hat{x}| \lesssim 1$) persists because stochastic timing still shifts the position of the velocity gradient even though its far-field amplitude is constrained. This scenario corresponds physically to a self-regulating rift in which crustal opening scales with the elapsed time since the previous event, a behaviour consistent with the observation that magmatic rifts accommodate most of the long-term plate divergence over geological timescales (Thordarson & Larsen 2007).

The central-limit interpretation is the following. A geodetic snapshot of the inter-rifting velocity field at any site within the deformation zone $x \lesssim aT$ encodes superposed contributions from dozens of past events. Stochastic scatter in individual T_j and b_j averages down over the ensemble of contributing events, and the CSS prediction converges to within the GPS measurement uncertainty for networks with aperture comparable to aT. Near-rift stations that respond primarily to the single most recent event retain the full cycle-to-cycle scatter shown in panel (a), and the CSS approximation is less accurate there; in that regime, knowledge of the specific rifting history is required to reduce the uncertainty.

4.5 Perspectives: the CSS as an inversion tool and its extension to heterogeneous rifts

The applications presented here treat T and b as known inputs and fit the single free parameter a to geodetic data. The CSS framework permits a more ambitious inversion. Because the velocity profile $v_{\mathrm{CSS}}(x, \tau)$ depends on three quantities (the deformation zone half-width aT, the far-field plate velocity b/T, and the cycle phase $\hat{\tau} = \tau/T$), a geodetic dataset that samples the near-rift gradient and the far-field plateau simultaneously constrains all three independently, with no locking depth or prescribed far-field forcing required. The gradient position fixes aT; the long-distance plateau fixes b/T; and their combination yields a and T individually, from which $\eta = EH/[2a(1-\nu^2)]$ follows once E and H are esti-

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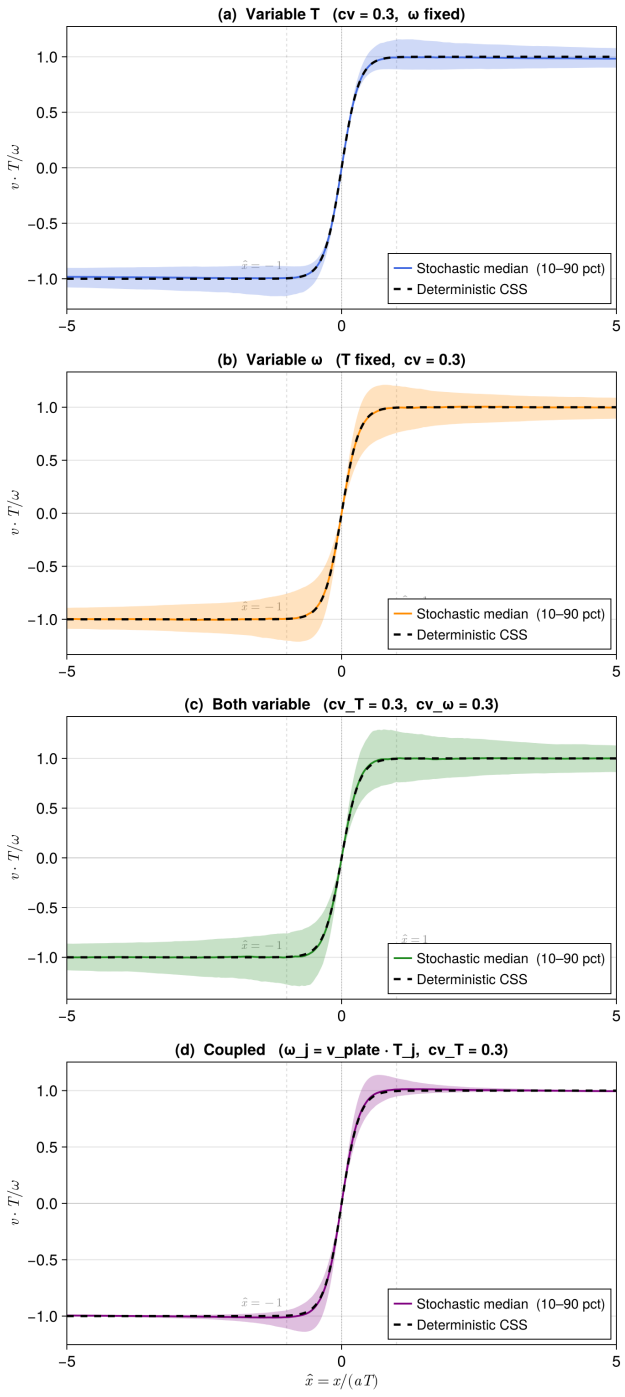


Figure 11. Robustness of the CSS velocity profile under stochastic recurrence ($a = T = b = 1$; phase $\hat{\tau} = 0.5$; 500 realisations at $c_v = 0.30$). Shaded bands show the 10th–90th percentile range; coloured lines are ensemble medians; the black dashed line is the deterministic CSS; vertical dashed lines mark $\hat{x} = \pm 1$. **(a)** Only T_j randomised. **(b)** Only b_j randomised. **(c)** Both T_j and b_j randomised independently. **(d)** Plate-rate-conserving coupling $b_j = v_{\text{plate}} T_j$. In all four cases the ensemble median coincides with the deterministic CSS; see text for the physical interpretation of the band shapes.

mated from seismological constraints. The stochastic analysis of Section 4.4 quantifies how cycle-to-cycle scatter in T_j and b_j propagates into uncertainty on these inferred parameters, providing a principled uncertainty budget for geodetically constrained viscosity estimates at any magmatic rift system.

Real rifts are segmented rather than homogeneous: dike intrusions span finite along-strike lengths, individual segments differ in recurrence history and plate thickness, and lateral variations in heat flow modulate the effective viscosity along the rift axis. The 1D framework developed here is immediately applicable to segmented geometries because its only length scale is the local product aT, set by the physical properties of each segment. Spatial variations in H or η along strike modulate aT locally without introducing new boundary conditions or requiring a finite model domain: each segment generates its own deformation halo, and the surface velocity field is their superposition. The stochastic ensemble of Section 4.4 already provides the natural language for this multi-segment setting: adjacent segments with different mean recurrence intervals T and variable event sizes b_j contribute independently to the observed velocity profile, and the CSS remains an unbiased predictor of the long-term average field as long as $\mathbb{E}[T_j] = T$ and $\mathbb{E}[b_j] = b$ hold locally for each segment. Extending the CSS to the full 2D problem (accounting for along-strike propagation of dike intrusions and rift segment interaction) is a natural next step, requiring the relaxation of the plane-strain approximation while retaining the same non-local Stokes coupling at the base of the elastic plate.

5 CONCLUSIONS

The CSS framework resolves the kinematic gap between post-diking transients and secular plate motion by recasting inter-rifting deformation as the time-averaged superposition of visco-elastic step responses to a long history of discrete dike intrusions. Because the model satisfies far-field plate divergence by construction and requires no locking depth, prescribed domain width, or external velocity offset, it provides a mechanically self-consistent alternative to backslip and dislocation models across any magmatic rift system.

At Krafla, the viscous step response fits the post-1984 differenced displacement record significantly better than the Foulger diffusion model ($\Delta\text{AIC} = -3.3$), yielding $a = 5.1 \times 10^{-5}$ m s^{-1} and an implied asthenospheric viscosity $\eta \approx 4 \times 10^{18}$ Pa s ($H = 10$ km, $E = 40$ GPa), consistent with post-glacial rebound estimates beneath Iceland and reflecting the anomalously hot sub-lithospheric mantle maintained by the Iceland plume. At Asal, statistical discrimination between the two models is inconclusive ($\Delta\text{AIC} = -0.84$); however, independent trilateration constraints on dike opening and on the early post-diking deformation rate both favour the viscous step model, which delivers a physically consistent best-fit $a = 9.5 \times 10^{-6}$ m s^{-1} and $\eta \approx 2.2 \times 10^{19}$ Pa s, a factor of five higher than at Krafla, in line with the lower heat flux of the Afar setting.

The non-local Stokes kernel underlying the viscous step model produces Lorentzian velocity tails that decay more slowly with distance than the erfc profile of Foulger-type diffusion. This structural difference is directly observable in

geodetic data: the viscous model outperforms the diffusion model in reproducing velocities at stations several kilometres from the rift axis, where the erfc profile underestimates the signal. The Krafla dataset provides the clearest discrimination; at Asal, a longer or denser geodetic transect would be required to separate the two formulations statistically.

Stochastic ensembles demonstrate that the CSS is an unbiased estimator of the long-term inter-rifting velocity field even at $c_v = 0.30$ cycle-to-cycle variability. Scatter in recurrence interval shifts the velocity gradient laterally, concentrating uncertainty in the transition zone; scatter in dike opening rescales the profile amplitude uniformly, leaving the gradient position unaffected. Coupling the two sources through the plate-rate constraint $b_j = v_{\mathrm{plate}} T_j$ eliminates far-field uncertainty entirely. In all cases the ensemble median coincides with the deterministic CSS, confirming that stochastic recurrence introduces spread but not bias.

These results establish the CSS as a practical tool for constraining sub-lithospheric viscosity from geodetic inter-rifting velocity profiles without auxiliary rheological assumptions. Jointly inverting the gradient position and far-field amplitude of a single velocity profile yields the deformation zone half-width aT, the plate velocity b/T, and the ratio EH/η simultaneously. Extending the framework to segmented rifts and to the full along-strike 2D geometry is the direct path toward a system-scale model of strain accumulation at magmatic plate boundaries, retaining the mechanically self-consistent structure derived here.

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Data Availability

The geodetic data used in this study are from published sources cited in the text. The analytical model code and inversion scripts are available at [PLACEHOLDER — repository DOI or URL upon acceptance].

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